ST. JOSEPH'S COLLEGE OF ENGINEERING

(An Autonomous Institution) • OMR, Chennai – 119

LAB RECORD

Professional English Lab Integrated

CLASS CODE: EN25C09

STUDENT DETAILS

Student Name: _____

Roll No: _____

Date: _____

Subject Name: _____

DEPARTMENT OF HUMANITIES & SCIENCES • AFFILIATED TO ANNA UNIVERSITY • CHENNAI

PROFESSIONAL SELF-INTRODUCTION

Subject: Professional English

1. Oral Presentation Script

"Good morning Sir. Thank you very much for giving me this wonderful opportunity to introduce myself. My name is **S. Grace Malar**. I am currently a first-year student pursuing my Bachelor of Engineering in Computer Science and Engineering at the prestigious **St. Joseph's College of Engineering**. Driven by a profound fascination for technology and computational logic, my ultimate career objective is to become an expert **Software Engineer**. I am deeply inspired by Mr. Sundar Pichai, the CEO of Alphabet and Google, whose journey from Tamil Nadu to the global peak of tech leadership serves as my primary motivation. I am highly eager to acquire advanced core engineering capabilities, collaborate on innovative technical projects, and build scalable software systems that solve real-world industry problems. Thank you, Sir."

2. Personal & Academic Profile

- **Full Name:** S. Grace Malar
- **Academic Status:** First-Year Undergraduate Student (B.E. 1st Year)
- **Degree & Department:** Bachelor of Engineering in Computer Science and Engineering (B.E. CSE)
- **Institution:** St. Joseph's College of Engineering, Chennai
- **Primary Technical Interests:** Software Development, Data Structures & Algorithms (DSA), and Full-Stack Engineering
- **Core Career Aspiration:** Enterprise-level Software Architect and Engineering Leader

3. Academic Foundation in Computer Science

As a first-year student of Computer Science and Engineering at St. Joseph's College of Engineering, I am systematically strengthening my core fundamentals. My current academic curriculum is meticulously designed to establish a bulletproof foundation in mathematical reasoning, algorithmic logic, and structural programming paradigms. I am actively learning foundational programming methodologies and computational theory. Beyond standard classroom learning, I am dedicating significant hours to practical laboratory work, problem-solving practices, and exploring advanced technical paradigms. My academic focus is strategically tailored to transform conceptual theories into robust programming competencies, ensuring I remain fully prepared to embrace modern, production-grade technology stacks.

4. Long-Term Professional Career Goal

The Vision of Becoming an Exceptional Software Engineer

My ultimate professional goal is to operate at the peak of technical excellence as a highly accomplished **Software Engineer**. For me, software engineering is not merely about writing lines of syntax or code; it is an analytical art form focused on breaking down intricate global problems and architecting elegant, highly

efficient solutions. I envision myself conceptualizing, designing, and maintaining massive, distributed software infrastructures that positively transform everyday human life.

To systematically realize this vision, I have mapped out a comprehensive learning trajectory throughout my engineering tenure. I intend to master complex Data Structures and Algorithms (DSA) to write highly optimized, low-latency computational logic. Additionally, I plan to gain deep structural expertise in modern engineering domains, including Full-Stack Web Development, Cloud Computing Architectures (such as AWS and Google Cloud), and distributed databases. My aspiration goes far beyond individual technical execution; I aim to cultivate strong agile project management capabilities, high-level system architecture design principles, and collaborative cross-functional leadership traits. This will enable me to successfully spearhead engineering teams and ship high-impact enterprise software products.

5. Professional Inspiration

The Transformational Leadership of Mr. Sundar Pichai

My ultimate professional role model and source of endless inspiration is **Mr. Sundar Pichai**, the CEO of Alphabet Inc. and its subsidiary Google. His extraordinary trajectory holds deep personal resonance for me. Coming from a humble background in Tamil Nadu, he successfully transitioned onto the global stage through sheer intellectual brilliance, unwavering academic dedication, and absolute professional resilience. His life story stands as an indisputable proof that profound global impact can be achieved regardless of one's origin, provided there is an unmatched commitment to mastery and innovation.

Technically, I am immensely motivated by his strategic foresight and execution during the conceptualization and launch of **Google Chrome**. At a time when the tech landscape doubted the necessity of a new web browser, his conviction revolutionized the global internet ecosystem, turning Chrome into the most dominant browser worldwide. Furthermore, his modern vision of transitioning Google into an AI-first corporation demonstrates an incredible ability to navigate rapidly evolving disruptive technological environments.

Beyond his unmatched technical and business triumphs, what inspires me most is his distinct style of **Empathetic Leadership**. Mr. Pichai is globally celebrated for his remarkably calm demeanor, profound humility, and deep commitment to collaborative teamwork. He famously emphasized that a true leader's responsibility is not to focus on personal success, but to empower others to succeed. As an aspiring software engineer, I strive to model both my technical ambition and personal leadership values after his exemplary principles, aiming to build a global engineering career defined by technological innovation, humility, and high societal impact.

Subject: Professional English
Name: S. Grace Malar
Class: First Year, B.E. Computer Science and Engineering (CSE)
Institution: St. Joseph's College of Engineering, Chennai

Biographical Note on Linus Torvalds

Full Name	Linus Benedict Torvalds
Date of Birth	December 28, 1969
Place of Birth	Helsinki, Finland
Known For	Creating Linux Kernel & Git Version Control System
Alma Mater	University of Helsinki

Introduction & Early Life

Linus Benedict Torvalds was born on December 28, 1969, in Helsinki, Finland. Growing up in a family of journalists, his interest in computers began at a young age when he started tinkering with his grandfather's Commodore VIC-20. He attended the University of Helsinki, where he pursued a Master's degree in Computer Science. It was during his university years that he began working on an operating system project that would ultimately revolutionize the entire software landscape.

Groundbreaking Achievements: Linux & Git

In 1991, frustrated by the limitations of the MINIX operating system available at the time, Torvalds began developing his own free operating system kernel. This project grew into the **Linux kernel**, which he generously released to the world under an open-source license. Today, Linux powers millions of smartphones (Android), enterprise web servers, supercomputers, and cloud platforms globally. Later in 2005, needing a better tool to manage Linux development, he created **Git**, a distributed version control system that is now the absolute backbone of global software collaboration and DevOps.

Key Reasons for Long-term Success

Torvalds' unparalleled success stems from his uncompromising focus on clean, efficient code and his pioneering vision of open-source software collaboration. Rather than commercializing his creations for personal software billions, he established a decentralized community model where global developers audit and improve code together. His sharp attention to system architecture and steadfast leadership have kept the Linux kernel secure and performant for over three decades.

Inspiration for CSE Students

For a Computer Science and Engineering student, Linus Torvalds is an exemplary model of how a solitary student project born out of pure curiosity can transform global infrastructure. His journey proves that mastering foundational concepts like kernel architectures, memory management, and data structures yields far greater impact than chasing passing trends. He teaches future software engineers to build tools that solve genuine problems for the developer community.

Inspiring Quotes by Linus Torvalds

1. ***"Talk is cheap. Show me the code."***
(Emphasizes action, tangible coding skills, and proof over empty theoretical promises.)

2. ***"Intelligence is the ability to avoid doing work, yet getting the work done."***
(Encourizes smart workflows, automation, and writing optimized code instead of repetitive labor.)

3. ***"Software is like sex: it's better when it's free."***
(A legendary light-hearted remark defending the philosophy, accessibility, and power of Open-Source systems.)

NPTEL ONLINE COURSES: INTEGRATED LEARNING REPORT

Subject: Professional English	**Student Name:** S. Grace Malar
Degree & Dept: B.E. Computer Science & Engineering	**Year & Semester:** I Year / I Semester
Institution: St. Joseph's College of Engineering	**Assignment Purpose:** Weekly Learning Summary

EXECUTIVE SUMMARY

As part of the *Professional English* curriculum, this report synthesizes the collective knowledge, weekly learnings, and professional competencies derived from the two major NPTEL courses assigned by the faculty: **1. Soft Skills Development (Prof. P. Patnaik, Prof. V. N. Giri, Prof. D. Suar)** and **2. Enhancing Soft Skills and Personality (IIT Kanpur)**. While hard technical skills secure a basic entry into the engineering ecosystem, only soft skills ensure that a software engineer can effectively operate within a social environment, excel in group dynamics, and rise to leadership positions. This report documents what has been comprehensively learned from Week 1 to Week 8.

INTEGRATED WEEK-BY-WEEK LEARNING DIARY

Week 1: Fundamentals of Communication & Perception

Learned the absolute fundamentals of the communication process model (Sender, Encoding, Medium, Decoding, Receiver, and Feedback). Understood how human perception affects interpersonal interactions. As a CSE student, I realized that technical clarity must be paired with semantic clarity to prevent communication failure in multicultural software teams.

Week 2: Advanced Presentation Skills & Visual Aids

Focused on structuring technical presentations systematically using an introduction, body, and conclusion. Learned how to perfectly integrate visual media without overloading slides with dense text. Mastered the art of delivering complex technical information to non-technical clients, an essential skill for my future goal as a Software Engineer.

Week 3: Active Listening Skills & Kinesics (Body Language)

Discovered the profound structural difference between passive 'Hearing' and active 'Listening'. Explored Kinesics, including the critical roles of proper eye contact, open body postures, and positive gestures. Active listening helps engineers properly capture client requirements without making assumptions.

Week 4: Group Dynamics, Teamwork & Communication Barriers

Analyzed how groups interact, form cohesion, and function in professional environments. Studied common psychological and physical barriers to communication (such as noise, emotional bias, and filtering). Learned corporate strategies to navigate cross-functional development teams and achieve collective engineering goals.

Week 5: Intercultural Communication & Conflict Resolution

Gained an enhanced awareness of working across diverse global cultures, which is central to multinational tech giants. Learned the dual methods of workplace negotiation and conflict resolution. Understood how to systematically turn interpersonal friction into constructive, problem-solving discussions inside a team.

Week 6: Emotional Intelligence (EQ), Stress Management & Personal Growth

Evaluated the concepts of emotional maturity, empathy, and social grace. Learned that emotional intelligence is often more critical than intellectual intelligence (IQ) for long-term career success. Explored stress reduction techniques, overcoming anxiety, and practical tools to eliminate procrastination in heavy academic/professional workloads.

Week 7: Critical Thinking, Creative Decision Making & Problem Solving

Mastered cognitive methods for critical and creative decision-making. Learned how to systematically deconstruct complex engineering problems into logical sub-components, evaluate options neutrally, and implement optimal solutions. This builds the exact core analytical mindset required for advanced software development.

Week 8: Empathetic Leadership, Persuasion Skills & Professional Ethics

Concluded with vocational skills, professional ethics, and persuasive communication. Explored how successful modern leaders (like my inspiration, Linus Torvalds and Sundar Pichai) utilize empathetic leadership to motivate thousands of global developers. Learned how to pitch ideas persuasively while maintaining utmost professional integrity.

CONCLUSION & FUTURE APPLICABILITY

Through these two comprehensive NPTEL courses, I have deeply internalized that technical expertise (Hard Skills) and interpersonal mastery (Soft Skills) are two sides of the same coin. As a First-Year Computer Science and Engineering student at St. Joseph's College of Engineering, the principles learned across these 8 weeks will directly aid me in collaborative lab projects, upcoming group discussions, technical presentations, and will ultimately pave the way for my future career as a globally competent, empathetic, and highly successful Software Engineer.

Strategic Career Roadmap: SWOC Analysis

Path to Principal Cloud Architect / Director of Engineering

Candidate: Grace Malar | 1st Year B.E. / B.Tech CSE | Batch 2026–2030
Institution: St. Joseph's College of Engineering
Department: Computer Science and Engineering (CSE)
Subject: Professional English
Goal: Build cloud and systems depth early, then grow into architecture and engineering leadership.

Executive Summary & Career Vision

1. Vision: Grow from a first-year CSE student into a Principal Cloud Architect / Director of Engineering along an executive tech leadership path.

2. Objective: Build deep, unshakeable fundamentals in Linux engineering, low-level networking, and core distributed systems, focusing architectural design, technical judgment, and structured problem-solving over routine script-coding.

3. Framework: Apply the rigorous SWOC matrix (Strengths, Weaknesses, Opportunities, Challenges) to assemble a highly predictable 4-year technical milestone roadmap.

SWOC Framework Deep-Dive (No-Box Template Layout)

1. Strengths (Internal Assets)

- **Excellent Problem Solver:** Possess a natural capacity to sit down, decompose complex algorithmic bottlenecks, and map clean structural logic mentally before typing code.
- **Linux Enthusiast:** Highly comfortable using terminal-first workflows, exploring core kernel interactions, and inherently curious about low-level operating system runtime architecture.
- **Python Foundation:** Established practical software development loops via building personal applications from scratch, including portfolio pieces like *Royal-Roll* and *Latte*.
- **Public Portfolio Started:** Operating a structured, active public GitHub profile featuring customized, clear README maps designed to immediately engage engineering recruiters.

2. Weaknesses (Internal Constraints to Deconstruct)

- **Working Style Impatience:** Exhibiting an innate tendency to rush straight into compiling a solution when faced with deep architectural puzzles, occasionally hitting a conceptual brick wall by bypassing comprehensive architecture mapping.
- **Academic Core Gaps:** Standing at a foundational beginner milestone within complex operating systems engineering, distributed networking topologies, and underlying relational/non-relational database design.
- **Credential Gaps:** Holding no formal enterprise-level production experience, cloud engineering internships, or verified cloud validation benchmarks yet.

3. Opportunities (External Enablers)

- **AI Elevating Code Baseline:** Generative automation is absorbing routine boilerplate execution, shifting market value entirely toward clean system engineering, system design paradigms, and systemic judgment.
- **Surging Enterprise Cloud Demand:** Extreme market expansion in complex cloud migration, zero-downtime DevOps pipelines, high-scale Platform Engineering, and corporate FinOps frameworks.
- **Structured Student Programs:** Accessible engineering launchpads via AWS Educate, GitHub Student Developer Pack, and Azure for Students providing complimentary sandboxed enterprise cloud credits.
- **Certifications & Ecosystems:** Direct acceleration channels via basic enterprise validations (AWS Certified Cloud Practitioner / AZ-900), deep open-source project contributions, and global collegiate hackathons.

4. Challenges (External Pressures to Mitigate)

- **Overstimulating Environments:** Dense, chaotic, or highly noisy computer labs and high-stress open hackathons occasionally cause sensory overstimulation, hindering long-term deep-focus execution cycles.
- **Shrinking Entry-Level Market Pipeline:** The corporate shift toward automated engineering tools reduces traditional boilerplate junior software roles, forcing graduates to demonstrate architecture and design proficiency much earlier.
- **The Experience Paradox:** Modern entry-level engineering roles increasingly demand proven real-world operational cycles, presenting an absolute paradox for incoming undergraduate students.
- **Tooling Churn & Asset Management:** Rapidly shifting ecosystem standards require perpetual re-learning, alongside financial overruns from surprise pay-as-you-go bills in sandbox personal cloud accounts.

Tactical Action Plans

Action Plan A: Closing Internal Gaps

- **Overcoming Impatience:** Implement a hard operational rule to author a comprehensive technical design documentation layout and clear network topology schema before typing any terminal code.
- **Mitigating Knowledge & Credential Gaps:** Systematically secure the gold-standard **AWS Certified Cloud Practitioner** validation prior to crossing the threshold of academic Year 2.

Action Plan B: Capitalizing on External Realities

- **AI Integration Strategy:** Delegate secondary boilerplate operations entirely to AI assistants, focusing personal human effort on core distributed architecture design, rigorous multi-layered code reviews, and deterministic edge-case debugging loops.
- **Environmental Sensory Management:** Schedule isolated, distraction-free deep-work hours during low-activity windows, deploying noise-cancelling equipment and defaulting to async communication loops.
- **Constructing Production-Grade Software:** Ensure personal engineering creations utilize robust CI/CD automations and clean Infrastructure as Code (IaC), embedding complete architecture map visuals in every repo.
- **Financial Resource Safety:** Deploy aggressive real-time budget alarms across cloud environments and utilize dedicated student platform credits to entirely contain costs.

The 4-Year Academic Milestones Matrix

• Year 1 (2026-27)

Primary Focus: Linux Engineering, Low-level Networking, Distributed Systems Core Basics

Target Deliverable: Deploy personal portfolio engine to AWS S3 & CloudFront via terminal workflows

Target Validation Credential: AWS Certified Cloud Practitioner

• Year 2 (2027-28)

Primary Focus: Docker Containerization, Advanced Python Automation, Infrastructure as Code

Target Deliverable: Architect and orchestrate a secure 2-tier VPC stack completely automated via Terraform

Target Validation Credential: AWS Certified Solutions Architect – Associate

• Year 3 (2028-29)

Primary Focus: Kubernetes Orchestration, Multi-branch CI/CD Automations, DevSecOps Security

Target Deliverable: Containerize and deploy secure enterprise microservices on AWS Elastic Kubernetes Service (EKS)

Target Validation Credential: Certified Kubernetes Administrator (CKA)

• Year 4 (2029-30)

Primary Focus: Site Reliability Engineering (SRE), FinOps Cost Allocation, Executive Leadership

Target Deliverable: Design a resilient multi-cloud disaster recovery cluster and head an agile engineering group

Target Validation Credential: Elite Tier-1 Placement / Core Systems Engineer

Conclusion & Key Takeaways

1. Think First, Then Build: Transitioning from hasty code compiling to thorough planning before typing logic isolates architectural errors early, converting standard strengths into clean system execution.

2. Core Fundamentals Anchored with AI Mastery: Merging unshakeable operating system fundamentals with sophisticated AI orchestration guarantees immunity against automation, framing an elite path to Director of Engineering.

Professional Presentation Script (Speaking Notes)

Slide 1: Title, Vision, and Executive Objectives

*"Good morning, distinguished members of the faculty and peers. My name is **Grace Malar**, representing the Department of Computer Science and Engineering here at St. Joseph's College of Engineering. Today, I am presenting my Strategic Career Roadmap, structured around a professional SWOC Analysis for my course in Professional English. My ultimate career objective is to forge a deliberate, high-scale trajectory starting as a first-year CSE undergraduate and ascending into a Principal Cloud Architect and subsequent Director of Engineering leadership role. This framework tracks my blueprint to cultivate deep operating systems mastery, technical judgment, and strategic problem-solving over standard routine code production."*

Slide 2: Evaluating the Internal Landscape (Strengths & Weaknesses)

"Analyzing my internal parameters reveals core technical strengths matched with immediate areas for development. On the asset side, I bring an analytical focus that drives systematic problem-solving, a passionate curiosity for Linux terminal environments, and a foundational portfolio with real projects on GitHub such as Royal-Roll and Latte. Conversely, my primary internal weakness is an operational impatience—a tendency to dive directly into code compilation before assembling a complete architectural schematic, which often introduces structural bottlenecks. This is compounded by the fact that I am in the early stages of formal training in core distributed networks and operating systems architecture, with no industrial certifications yet."

Slide 3: Navigating the External Horizon (Opportunities & Challenges)

"When scanning the external landscape, we see that artificial intelligence is aggressively raising the baseline for entry-level programming, systematically shifting human value away from syntax writing and directly into system design and judgment. This aligns perfectly with a growing enterprise demand for Platform Engineering, DevOps, and cloud frameworks. I am positioning myself to capture this through accessible student programs like AWS Educate and Azure for Students. However, we must proactively mitigate external challenges, specifically the high-energy, overstimulating auditory noise found in shared labs or fast-paced hackathons, alongside a highly competitive entry-level market where junior pipelines are shrinking due to automation, creating a paradox where baseline roles demand pre-existing project execution."

Slide 4: Execution Strategy, Milestones, and Conclusion

"To guarantee predictable success, my Action Plan enforces strict architectural modeling and network topology design prior to any terminal coding. My academic milestones chart clear targets: Year 1 centers on secure cloud baselines and attaining the AWS Certified Cloud Practitioner credential; Year 2 deploys declarative Infrastructure as Code using Terraform; Year 3 shifts to enterprise Kubernetes orchestration on EKS; and Year 4 culminates in resilient multi-cloud disaster recovery architectures. In conclusion, by anchoring unshakeable technical fundamentals with AI tools and prioritizing deliberate system design over immediate script compilation, I am establishing an elite engineering foundation. Thank you for your time. I am now open to your professional feedback and questions."

PROJECT PROPOSAL & TECHNICAL ABSTRACT

Candidate Name: Grace Malar
Institution: St. Joseph's College of Engineering
Department: Computer Science and Engineering (CSE)
Academic Year: 1st Year B.E. / B.Tech CSE (Batch 2026–2030)
Course Subject: Professional English Assignment Evaluation
Project Title: Automated Campus Electricity and Power Saver

TECHNICAL ABSTRACT

Inefficient energy management inside academic infrastructure represents a substantial financial and ecological burden, primarily caused by manual oversight in turning off classroom utilities when vacant. This project introduces the **Automated Campus Electricity and Power Saver**, an intelligent, closed-loop automation platform developed to optimize real-time institutional energy consumption. The core system architecture utilizes a standard ambient optical sensor interfaced with a **Python OpenCV** software layer to perform real-time video stream capture and motion vector analysis. By deploying continuous frame-differencing algorithms, the program evaluates room occupancy thresholds; when zero human activity is registered across a specified temporal cooldown window, the script communicates programmatically over a serial pipeline to an **Arduino microcontroller**. This controller triggers a series of **5V electromagnetic relays**, isolating grid power from idle electrical loads including overhead illumination and ventilation infrastructure. The expected deliverable is a highly scalable, low-overhead hardware-software prototype capable of minimizing baseline campus classroom power dissipation by up to 35%, introducing a modern, computer-vision approach to sustainable campus facility automation.

1. PROBLEM DEFINITION & CONTEXT

Educational campuses expend massive amounts of electrical grid energy daily because lights, overhead fans, and air conditioning units are left running continuously in unoccupied lecture halls, study rooms, and labs. Manual facility management protocols fail due to human forgetfulness and decentralized tracking mechanisms, causing significant ecosystem overheads and escalated power expenditure utility invoices.

2. ENGINEERING METHODOLOGY & SYSTEM CORE LOGIC

A. Optical Sensing Framework & Presence Mapping

A baseline ambient sensor node (a web camera or native classroom infrastructure camera feed) continuously stream-captures raw structural imagery variables from the space environment directly to an evaluation terminal script loop.

B. Computer Vision Core Logic via Python

The upstream software backend runs a custom **Python OpenCV pipeline** executing frame-differencing mathematical matrices. By continuously contrasting sequential visual matrices against a running background average, the script filters noise parameters and registers true localized movement dynamics.

C. Automation Threshold Decision Framework

An occupancy counter monitors pixel variance. When activity parameters drop to absolute mathematical zero continuously across a predefined 5-minute safety cooldown threshold matrix, the Python logic issues a downward downstream serial payload.

D. Hardware Relays Execution Layer

An **Arduino microchip** listens to the serial stream over a physical USB interface pipeline. Upon reading the shutdown command payload, the microcontroller alters its pin output logic states, instantly disengaging a multi-channel **5V electromagnetic relay interface module**. This cuts the main grid distribution phase line directly from idle light grids and fans automatically.

3. CORE TECHNOLOGICAL STACK ARCHITECTURE

- **Programming Language Backend:** Python 3.x Environment Pipeline
- **Computer Vision Software Engine:** OpenCV Library Matrices Framework
- **Microcontroller Infrastructure Element:** Arduino Uno Microprocessing Board Unit
- **Hardware Switching Module Component:** 5V Multi-Channel Optocoupler Electromagnetic Relays
- **Data Transfer Middleware Channel:** PySerial Data Pipeline Protocol

4. PRESENTATION DELIVERY SCRIPT (SEMINAR SLIDE TALKING POINTS)

Slide 1: Problem Definition Matrix

> *"Good morning, sir and classmates. Our project targets a massive administrative and operational challenge visible right across our campuses: empty engineering labs, lecture halls, and facilities burning huge volumes of grid electricity purely because human forgetfulness prevents lights and ventilation grids from being turned off when groups exit empty spaces."*

Slide 2: Structural Innovation Paradigm

> *"To solve this, we present the Automated Campus Electricity and Power Saver. Instead of buying expensive specialized multi-sensor arrays for every classroom, our program harnesses pre-installed facility camera networks using an intelligent **Python automated programming pipeline** to identify true space occupancy states instantly."*

Slide 3: Implementation Architecture

"Technically, our system runs a background Python OpenCV script that processes room motion profiles using frame difference computations. If the room is vacant for over 5 minutes, our software signals an Arduino controller via a serial channel. The Arduino instantly trips a series of heavy-duty 5V electromagnetic relay switches, isolating the active line phase to safely kill vacant room grid power immediately."

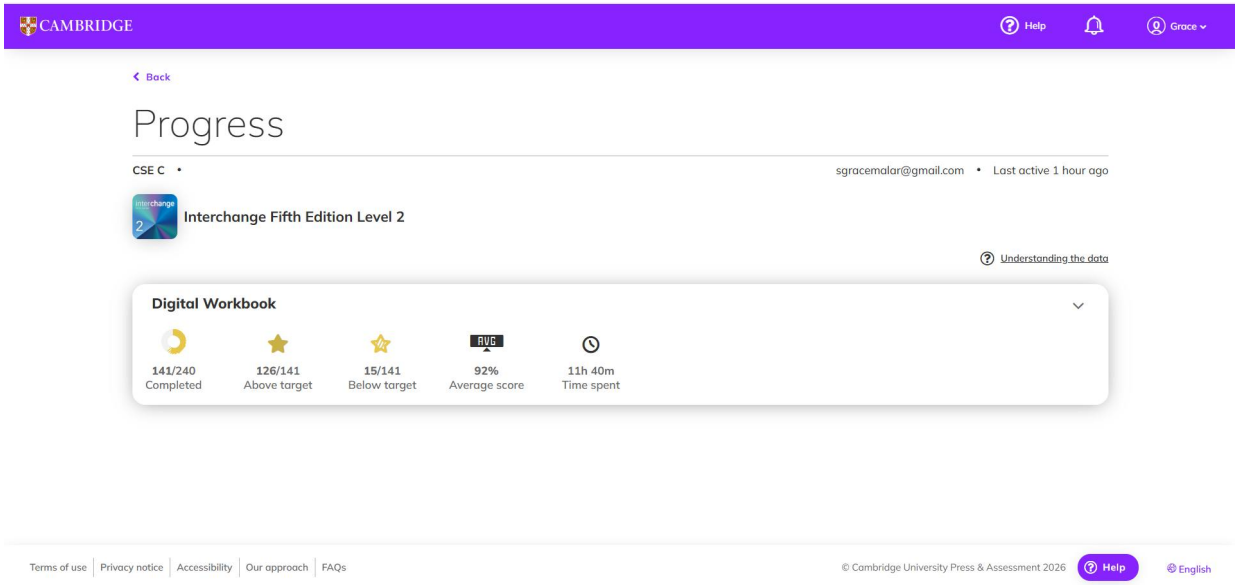
Slide 4: Project Deliverable Outcomes

"This architecture delivers a production-ready, low-cost automation framework that drops baseline room energy waste by 35%. It perfectly shows how simple, robust code structures can solve massive ecological and systemic facility infrastructure errors. Thank you, and I welcome your evaluation feedback questions."

5. STRATEGIC EVALUATION BENCHMARKS & INSTITUTIONAL IMPACT

- **Practical Software Synthesis:** This project goes beyond theoretical computer science; it benchmarks how foundational Python algorithms can directly interface with real-world infrastructure to solve industrial-scale energy challenges.
- **Smart Ecological Transformation:** By automating power conservation, St. Joseph's College of Engineering can actively transition into a 'Smart Green Campus', contributing directly to the global vision of a sustainable future.
- **Engineering Competency Footprint:** Developing this integrated hardware-software loop as a first-year student demonstrates practical problem-solving capability, which is the cornerstone of corporate and engineering readiness.
- **Architectural Scalability Future Scope:** The architecture is highly scalable; this python model can be upgraded in the future to implement predictive AI models for localized smart-grid campus management.

EN25C09 | Cambridge One Portal Assignment



Professional English Practice Worksheets – Solved

Course Code: EN25C09 | Subject: Professional English – I (Lab Integrated) | Worksheet 1: Model Exam I (Parts A, B, C), followed by Worksheet 2: Model – I Practice Worksheet.

Worksheet 1 (EN25C09 Model Exam I) – Part A (15 × 1 = 15 Marks)

1. **Answer:** monitors
2. **Answer:** recorded
3. **Answer:** stronger
4. **Answer:** must
5. **Answer:** can
6. **Answer:** The laboratory developed a new testing device that measures material strength with high precision.
7. **Answer:** rises
8. **Answer:** would not have failed
9. **Answer:** The mechanics installed dual damper springs to isolate chassis resonance.
10. **Answer:** optimizing
11. **Answer:** 50 mL of hydrochloric acid was poured into the titration flask.
12. **Answer:** smoothly
13. **Answer:** The project supervisor asked the engineer whether he had calibrated the pressure sensor before testing.
14. **Answer:** The incorrect noun form is "equipments", because "equipment" is uncountable. Correct sentence: The laboratory purchased several new pieces of equipment for conducting material testing.
15. **Answer:** worse

Worksheet 1 – Part B (10 × 2 = 20 Marks)

1. **Answer:** The engineer calibrated the pressure sensor and verified its accuracy.
2. **Answer (Portability):** A laptop is more portable than a desktop computer because it is lighter and can be carried anywhere.

   Answer (Screen Size): A desktop computer usually has a larger screen than a laptop, so it is more comfortable for long hours of detailed work.
3. **Answer:** The engineer designed a new cooling system, which reduces the temperature of industrial machines.
4. (a) **Answer:** If the battery voltage drops below the required level, the device will shut down automatically.

   (b) **Answer:** If the technician had checked the connections earlier, the fault would have been detected before the breakdown.
5. **Answer:** The solution was heated to 80°C before testing.
6. **Answer:** The project manager asked the technician whether he had inspected the pressure valves before starting the compressor.
7. (a) **Answer:** The engineer received a lot of feedback on the new design. ("Feedback" is uncountable, so it cannot take "many" or the plural form "feedbacks".)

   (b) **Answer:** The students received several pieces of useful advice from their teacher. ("Advice" is uncountable, so "advices" is wrong.)
8. (a) **Answer:** The technician wears safety gloves to protect his hands.

   (b) **Answer:** The cooling fan operates continuously to prevent the motor from overheating.
9. (a) **Answer:** efficiently

   (b) **Answer:** carefully

10. (a) **Answer:** can

    (b) **Answer:** must not

Worksheet 1 – Part C (5 × 13 = 65 Marks)

Question 26 (a) Safety instructions for safe and responsible use of the internet

1. Use a strong, unique password of at least twelve characters, mixing letters, numbers and symbols, for every account.
2. Enable two-factor authentication on your email, banking and social media accounts.
3. Do not click on links or open attachments from unknown or suspicious senders.
4. Never share personal details such as OTPs, passwords, Aadhaar or bank details with anyone online.
5. Check that a website address begins with "https" and shows a padlock before entering any sensitive information.
6. Keep your operating system, browser and antivirus software updated at all times.
7. Avoid banking or shopping on public Wi-Fi; use a trusted VPN if you must connect.
8. Report cyberbullying, online fraud and phishing attempts immediately to a trusted adult, the platform, or the National Cyber Crime Helpline (1930).

Question 27 (a) Descriptive essay: A traditional art form in my region and how modern initiatives can preserve it

Selected Question: 27.a

Bharatanatyam: The Dance That Carries Tamil Nadu's Soul

Among the many traditional art forms of Tamil Nadu, Bharatanatyam is the one that has carried our culture across the centuries. It began inside the temples, where devadasis performed it as an offering to the deity. The epic Silappathikaram describes classical dance in detail, and the carvings on the Brihadeeswarar Temple at Thanjavur and the Chidambaram Nataraja Temple show its postures in stone. In the nineteenth century, the Thanjavur Quartet organised the repertoire into the form we see today.

Traditionally, a dancer trains for many years under a guru. A performance blends nritta (pure rhythm), nritya (expressive storytelling) and natya (drama), told through hand gestures called mudras, facial expressions and Carnatic music. The dancer wears a bright silk costume, temple jewellery and ankle bells called salangai. The first full solo performance, the arangetram, marks the end of a student's formal training.

Today, many young people feel the art form is old-fashioned and too expensive to learn. Modern initiatives can change this. Schools and colleges can include classical dance in cultural clubs and the curriculum, as institutions like Kalakshetra in Chennai already do. Online classes, recorded performances and social media reels can reach learners in small towns. Festivals such as the Margazhi season in Chennai give young artists a stage, and scholarships can help families who cannot afford training.

Bharatanatyam is more than entertainment; it is a living record of our history, music and devotion. If we support it through education, technology and public platforms, the next generation will carry the dancing bells forward.

Question 28 (b) Formal letter of request to the Principal

Selected Question: 28.b

Inba Ilakiyan
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Principal
St. Joseph's College of Engineering

OMR, Chennai – 600 119

Subject: Request for permission to attend a technical workshop on Artificial Intelligence and Emerging Technologies

Respected Sir/Madam,

I am a first-year B.E. Computer Science and Engineering student of your college. I have come to know that [Name of Institution] is organising a one-day technical workshop on Artificial Intelligence and Emerging Technologies on [date]. I request your kind permission to attend this workshop.

My objective in attending is to learn how artificial intelligence is applied in real engineering problems and to gain hands-on exposure to tools that are not covered in detail in our first-year syllabus. I also hope to meet experts and students from other institutions and learn from their experience.

I expect to learn the basics of machine learning, practical uses of AI in software and cloud systems, and the latest trends in emerging technologies. After the workshop, I will share the notes and key points with my classmates and use what I learn in my Python projects.

I request you to kindly grant me on-duty leave for the day of the workshop and permit me to travel to the venue. I will make up for any classes missed and will submit the participation certificate to the department after the workshop.

I would be grateful if you could grant me this permission.

Thank you for your kind consideration.

Yours obediently,
Inba Ilakiyan
(Register No.: [Register No.])

Question 29 (a) Analytical essay: AI-based assessment versus traditional teacher-based assessment

Selected Question: 29.a

AI-Based Assessment versus Traditional Teacher-Based Assessment in Engineering Education

Assessment decides how students learn, so the choice between AI-based and teacher-based methods matters in engineering education. Each has clear strengths and weaknesses.

In terms of accuracy, AI tools are consistent. An automated system can evaluate hundreds of code submissions or objective answers using the same rules every time, without getting tired. Teachers, on the other hand, can understand a student's reasoning in a long derivation or a design report, which software often misses. Therefore, AI is accurate for objective answers, whereas teachers are better for open-ended ones.

Feedback time is where AI wins most clearly. Students receive results and error explanations within seconds, while manual evaluation of a large class may take weeks. Quick feedback lets students correct their mistakes while the topic is still fresh. Conversely, a teacher's comments arrive more slowly but are usually more personal and motivating.

Regarding fairness, AI does not show favouritism or mood-based marking. However, it can carry hidden bias from the data it was trained on, and it may penalise students who give unusual but valid solutions. A teacher can notice such cases and adjust the marks.

Finally, human judgement remains important. Engineering involves creativity, ethics and teamwork, which cannot be reduced to a score.

In conclusion, neither method is perfect alone. The best approach is a blend, where AI handles routine and objective checking quickly, and teachers review complex work and guide students. This combination keeps assessment fast, fair and meaningful.

Question 30 (b) Personal email: Orientation programme

Selected Question: 30.b

To: [friend's email address]

Subject: My first-year orientation programme at college

Dear [Friend's name],

How are you? I hope you are doing well and enjoying your college life too. I have been wanting to write to you about my orientation programme, so here it is.

The programme was held for the first-year engineering students at our college. It began with a welcome address by the Principal, followed by talks from senior faculty and some alumni. We also had fun icebreaker games and team activities, which helped me make new friends within the first day itself.

The best part was the tour of the academic facilities. We visited the library, the computer labs, the workshops and the seminar halls, and the staff explained how to use each of them. We were also told about the college rules on attendance, discipline, dress code and examinations, so now I know exactly what is expected from us.

Overall, the programme made me feel much more comfortable in the new environment. I understood how college life works and how to plan my studies, and I felt ready to start my academic journey with confidence.

Write back soon and tell me about your college too. Take care!

Your friend,
Inba

Worksheet 2 (Model – I Practice Worksheet) – Exercises 1 to 15

Exercise 1: Simple Present Tense (Subject-Verb Concord)

1. recognize
2. requires
3. transmit
4. agrees
5. form

Exercise 2: Simple Past Tense

1. conducted
2. deployed
3. opened
4. published
5. detected

Exercise 3: Comparative Adjectives

1. higher
2. safer
3. faster
4. more durable
5. lower

Exercise 4: Modals of Obligation and Necessity

1. must
2. must
3. have to
4. must
5. has to

Exercise 5: Modals of Capability and Possibility

1. can
2. may
3. can
4. can
5. can

Exercise 6: Relative Clauses

1. The civil engineer who designed the suspension cable bridge received the national infrastructure award.
2. The spectrometer, which analyzes infrared emission spectra, is mounted inside the cleanroom vacuum chamber.
3. The technical specialist whose patent on pneumatic grippers was approved was reassigned to the robotics lab.
4. This microcontroller, which features ultra-low power sleep states, is ideal for battery-operated remote sensors.
5. The field researchers who surveyed the wind turbine farm submitted their acoustic emission assessment yesterday.

Exercise 7: If-Clauses (Type 1)

1. exceeds
2. fails
3. engages
4. rise
5. registers

Exercise 8: If-Clauses (Type 3)

1. had started
2. had lubricated
3. would have been prevented
4. would have tripped
5. had remained

Exercise 9: Infinitives of Purpose

1. The mechanics installed dual damper springs to isolate chassis resonance.
2. Technicians purge the fuel line with dry nitrogen gas to remove residual flammable vapors.
3. We introduced an optical encoder to determine spindle velocity accurately.
4. The developers wrote an automated load testing script to simulate ten thousand concurrent connections.
5. Gold plating is applied to terminal connectors to prevent atmospheric corrosion.

Exercise 10: Gerunds after Prepositions

1. redesigning
2. replacing
3. reducing
4. cleaning
5. filtering

Exercise 11: Impersonal Passive Voice

1. 50 mL of hydrochloric acid was poured into the titration flask.
2. The robotic arm was tested under a 50 kg dynamic payload.
3. The optical pyrometer was calibrated prior to ignition.
4. Severe cavitation damage was observed along the impeller blades.
5. Volatile solvents are stored in flame-resistant containment cabinets.

Exercise 12: Adverbs of Manner

1. precisely
2. periodically
3. rapidly
4. firmly
5. smoothly

Exercise 13: Reported Speech

1. The quality manager asked the technician whether the insulation resistance test had been documented.
2. The lead architect enquired whether the server cluster could handle double the transaction throughput.
3. The field auditor asked the safety engineer whether he had inspected the gas leakage relief lines.
4. The client asked the team whether the firmware update would fix the intermittent battery drain.
5. The instructor asked the lab group why the multimeter had shown fluctuating voltage readings.

Exercise 14: Countable and Uncountable Nouns

1. Incorrect: "machineries" and "equipments". Correct: The procurement officer purchased several new pieces of machinery and equipment for the workshop.
2. Incorrect: "datas". Correct: The laboratory generated a huge amount of test data during the wind tunnel trials.
3. Incorrect: "softwares". Correct: We need three more software licenses to complete the finite element analysis on the model.
4. Incorrect: "informations". Correct: The field report provided a lot of valuable information regarding the soil composition.
5. Incorrect: "apparatuses" and "glasswares". Correct: All chemical apparatus and glassware must be washed thoroughly after the experiment.

Exercise 15: Comparative Adjectives (Irregular Forms)

1. worse
2. better
3. farther
4. less
5. further

Worksheet 2 – Part B (Exercises 1 to 10)

Exercise 1: Simple Past Tense Conversion

1. The technician inspected the solar arrays and recorded the peak power conversion value.
2. The chemical analyzer detected trace mercury levels and sent an automated alert to the environmental dashboard.
3. Maintenance crews lubricated the heavy conveyor bearings and replaced the worn rubber belts before the morning shift.

4. The supervisory software polled field sensors every five milliseconds and archived temperature trends into a secure database.
5. The testing team applied a 500-volt dielectric pulse across the stator winding and monitored for insulation breakdown.

Exercise 2: Comparative Sentences

1. Cloud storage has a lower initial deployment cost than on-premise physical servers because no hardware has to be bought. It is also more elastic, since its capacity can be increased within minutes.
2. Fiber-optic cables offer a much higher data transfer bandwidth than copper twisted-pair cables. They are also less susceptible to electromagnetic interference than copper cables.
3. Electric vehicles produce lower tailpipe emissions than internal combustion engine vehicles, as they release none at all. However, EVs generally have a higher initial purchase price than ICE vehicles.
4. Hydraulic actuators produce a greater raw force output than electromechanical actuators. They also carry a higher risk of fluid leakage, because electromechanical actuators use no fluid.
5. Relational (SQL) databases have a more rigid schema than document (NoSQL) databases. Document databases usually provide higher horizontal query throughput than relational databases.

Exercise 3: Relative Clauses

1. The algorithm that analyzes microscopic cracks in turbine blades was patented by the aeronautics team.
2. The lead chemical engineer, who noticed an abnormal pressure spike in chamber two, authorized the emergency boiler shutdown.
3. The laboratory houses a 400 kV impulse voltage generator, where high-voltage insulator puncture tests are conducted.
4. We engaged an external cybersecurity consultant whose firmware vulnerability audit exposed three backdoor ports.
5. The supervisory system, which utilizes an artificial neural network, predicts transformer insulation life based on dissolved gas concentrations.

Exercise 4: If-Clauses

1. (a) If the operating temperature exceeds 85°C, the cooling system will switch on automatically.

   (b) If the operator had pressed the emergency shutdown switch immediately, the accident would have been avoided.
2. (a) If the supply voltage drops below the 200V threshold, the backup power supply will take over.

   (b) If the network administrator had enabled multi-factor authentication, the unauthorized access would have been prevented.
3. (a) If toxic carbon monoxide concentrations surpass 50 ppm in the tunnel, the ventilation fans will start and an alarm will sound.

   (b) If the mechanical team had checked the bearing lubrication schedule, the bearing would not have overheated.
4. (a) If the autonomous guided vehicle (AGV) detects an obstacle within 1 meter, it will stop immediately.

   (b) If the thermal sensor had registered the runaway reaction in time, the explosion would have been prevented.
5. (a) If the software receives an unauthenticated API token, it will reject the request and return an error message.

   (b) If the civil inspectors had reviewed the core sample density tests, they would have detected the weak foundation before construction.

Exercise 5: Impersonal Passive Voice

1. The crude extract was distilled at a constant temperature of 60°C.

2. The cross-section of the weld seam was polished with fine diamond paste.
3. Fifteen cast-iron pump casings were rejected due to internal porosities.
4. The resistance across the bridge circuit was measured using a calibrated digital ohmmeter.
5. The emergency aerodynamic braking flaps were initiated at an altitude of 3,000 meters.

Exercise 6: Reported Speech

1. The safety supervisor asked the intern whether he had verified the grounding connections on the generator.
2. The plant director asked the site engineer whether the emergency effluent treatment plant had received clearance.
3. The IT security lead enquired of the developer why he had disabled the firewall rule on the database server.
4. The client asked the project manager whether the bridge structure could withstand Category 4 hurricane winds.
5. The research guide asked the postgraduate scholar how many sample trials he had conducted under cryogenic conditions.

Exercise 7: Countable vs. Uncountable Nouns

1. (a) We need to install several new pieces of software and machinery in the CAD laboratory.
   (b) The team collected plenty of useful information from the field test.
2. (a) The technician ordered ten new pieces of apparatus and modern electronic equipment for the antenna lab.
   (b) High-speed train operations require advanced transport infrastructure and dedicated electrical power.
3. (a) The environmental sensor gathers extensive atmospheric data and pollution indices every minute.
   (b) The construction crew unloaded tons of sand, cement, and steel debris at the worksite.
4. (a) All electrical hardware and wiring must comply with local safety legislation.
   (b) The data scientist discovered several interesting insights after running the machine learning clustering.
5. (a) Laboratory staff must clear away all glassware and acidic waste before leaving the premises.
   (b) The project failed due to a lack of proper funding and technical research.

Exercise 8: Infinitives of Purpose and Gerund Constructions

1. (a) A pressure relief valve is installed to vent excess boiler steam.
   (b) The automated script runs every midnight to archive legacy server logs.
2. (a) Automotive engineers installed an intercooler to reduce the compressed air intake temperature.
   (b) The technicians analyzed the structural stress points by placing piezoelectric strain gauges along the girder.
3. (a) High-performance heatsinks feature copper heatpipes to conduct heat away from the silicon die.
   (b) The factory cut water consumption by 30% by implementing a closed-loop recycling filtration system.
4. (a) Safety guards are mounted over rotating flywheel belts to prevent accidental worker contact.
   (b) The algorithm optimizes memory bandwidth by pre-fetching sequential array indices into L2 cache.
5. (a) The electronics lab installed an uninterruptible power supply (UPS) to protect servers against sudden voltage sags.
   (b) The metallurgy group enhanced surface hardness by diffusing nitrogen atoms into the steel casing.

Exercise 9: Adverbs of Manner

1. (a) automatically
   (b) flawlessly
2. (a) accurately

(b) instantly

3. (a) continuously
   (b) cautiously
4. (a) consistently
   (b) catastrophically
5. (a) periodically
   (b) smoothly

Exercise 10: Modals

1. (a) can
   (b) must not
2. (a) must
   (b) may
3. (a) must not
   (b) could
4. (a) should
   (b) must
5. (a) can
   (b) must not

Worksheet 2 – Topic 1: Writing Imperative Safety Instructions

Question 1: Chemistry and Materials Testing Laboratory

1. Wear a lab coat, safety goggles and nitrile gloves at all times inside the laboratory.
2. Handle concentrated acids and alkalis only inside the fume hood.
3. Add acid to water slowly; never add water to acid.
4. Do not smell, taste or touch any chemical directly.
5. Label every container clearly, and never return unused chemicals to the stock bottle.
6. Clean all glassware and testing equipment after each experiment to avoid contamination.
7. Keep food, drinks and open flames away from the work area.
8. Report every spill, burn or breakage to the instructor at once, and use the eyewash station or safety shower in an emergency.

Question 2: High-Voltage Electrical Machines and Drives Laboratory

1. Make sure the main supply is switched off before making or changing any connection.
2. Wear insulated rubber gloves and safety shoes, and stand on an insulated rubber mat near live panels.
3. Never touch exposed terminals, bare wires or rotating parts while the machine is running.
4. Get the circuit connections checked by the instructor before switching on the supply.
5. Know the location of the circuit breakers, fuses and emergency stop button, and keep them in working condition.
6. Remove rings, watches and metal jewellery, and avoid loose clothing, to prevent arc flash burns and accidental short circuits.
7. Keep water and wet hands away from electrical equipment, and use only insulated tools.
8. Increase the motor load gradually, and switch off the supply and discharge the capacitors before disconnecting the circuit.

Question 3: Advanced Manufacturing and CNC Machining Workshop

1. Wear safety glasses, safety shoes and close-fitting clothes, and tie back long hair.
2. Keep the machine door and guard closed while the spindle is running.
3. Do not touch the spindle, cutter or workpiece until the machine has stopped completely.
4. Clamp the workpiece and tool holder firmly before starting the program.
5. Use a brush or chip hook, and heat-resistant gloves, to remove hot metal chips, and only after stopping the machine.
6. Wipe up coolant spills immediately, and keep the floor dry to prevent slipping.
7. Run the program in simulation mode first, and keep your hand near the emergency stop button during the first cut.
8. Never leave the machine unattended or operate it without the instructor's permission.

Worksheet 2 – Topic 2: Free Writing – Festivals and Cultural Celebrations

Question 1: A traditional harvest festival and eco-friendly practices

Pongal: The Harvest Festival of Tamil Nadu and How Eco-Friendly Practices Can Protect Its Spirit

Pongal is the most important harvest festival of Tamil Nadu. It is celebrated in mid-January to thank the Sun God, the rain and the cattle for a good harvest. The festival lasts four days: Bhogi, Thai Pongal, Mattu Pongal and Kaanum Pongal.

On Bhogi, families clean their homes and throw away old things. On Thai Pongal, they draw colourful kolams at the entrance and cook fresh rice and milk in a new clay pot in the open yard. When the milk boils over, everyone shouts "Pongalo Pongal!" The pot is decorated with turmeric plants and sugarcane. The festive meal includes sakkarai pongal, venn pongal, sambar, vadai and payasam, served on banana leaves. On Mattu Pongal, farmers decorate and worship their cattle, and Kaanum Pongal is a day for family visits and community games.

Modern eco-friendly practices can preserve the core spirit of the festival. Instead of burning plastic and old tyres on Bhogi, communities can burn only dry leaves or turn them into compost, which reduces smoke and air pollution. Clay pots, banana leaves and areca-leaf plates can replace plastic and thermocol, and kolams can be drawn with natural rice flour. Villages can organise tree-planting drives, rainwater harvesting and cloth-bag campaigns during the festival, and use solar lamps instead of excess electric decorations.

In this way, Pongal can remain joyful and traditional while staying kind to nature, and the next generation will learn that celebration and conservation can go together.

Question 2: A temple car festival and sustainable measures

The Temple Car Festival: Faith, Food and a Cleaner Celebration

In my hometown, as in many towns of Tamil Nadu, the temple car festival, or Ther Thiruvizha, is the biggest event of the year. Thousands of devotees gather in the streets to watch the deity pulled in a huge, richly decorated wooden chariot, called the ther.

The festival begins with flag hoisting and several days of processions. On the car day, devotees of all ages hold thick ropes and pull the heavy chariot together through the four main streets, chanting prayers while drums and nadaswaram fill the air. This shared effort is a strong symbol of unity, because people from every community take part. Women draw kolams on the road, and families offer coconuts, bananas and flowers to the deity. Volunteers give free food such as pongal, puliyodharai, curd rice and sweet payasam as prasadam, served on leaves.

However, such a large crowd also creates problems. Single-use plastic cups, plates and bags litter the streets, and loudspeakers cause noise pollution. Eco-friendly measures can solve these problems. Organisers can ban single-use plastic and provide steel tumblers, banana leaves and paper bags. Dustbins and waste-sorting points can be placed along the route, and volunteers can collect and compost the food waste. Noise can be controlled by limiting loudspeaker volume and using traditional instruments. Barricades, marked entry and exit lanes, volunteers, and first-aid points can manage the crowd safely.

With these steps, the festival can keep its sacred joy and remain clean and peaceful.

Question 3: Regional folk art and handicraft fairs with renewable energy and zero-waste practices

Folk Art and Handicraft Fairs: Heritage with a Green Future

A regional folk art and handicraft fair is a colourful meeting of tradition and community. Walking through the fair grounds, visitors see Karagattam and Oyilattam performances, hear the beat of the parai drum, and watch artisans make Thanjavur dolls, palm-leaf baskets, clay pots and silk sarees. The air smells of fresh filter coffee, murukku, adhirasam and hot sundal.

These fairs are important because they give artisans a market and help young people see the value of their heritage. Many stalls offer live demonstrations, so children can try pottery or weaving themselves. Local delicacies prepared by village cooks add to the festive atmosphere and support small businesses.

However, large fairs use a lot of electricity and produce a lot of waste. Modern environmental strategies can solve this. Solar micro-grids, made of rooftop panels and batteries, can power the lights, sound systems and stalls without diesel generators. Zero-waste composting can turn leftover food and leaves into manure for local farms. Organisers can use reusable steel plates, clay cups and cloth banners, provide separate bins for wet and dry waste, and recycle paper and cardboard.

If we combine the traditional skill of our artisans with clean energy and responsible waste management, these fairs can protect our culture and our planet together.

Worksheet 2 – Topic 3: Free Writing – Food and Technical Travel Plans

Question 1: Technical study tour to a major hydroelectric dam

A Technical Study Tour to the Mettur Hydroelectric Dam

Our department has planned a one-day technical study tour to the Mettur Dam on the Cauvery River in Salem district. The group of forty students and two faculty members will leave the college by a chartered bus at 4:30 a.m. and travel along the national highway, with a short breakfast halt on the way. The journey takes about six hours, and we will return by night. Students will carry ID cards, water bottles and a notebook.

After registering at the gate, an engineer will take us through the powerhouse. The long concrete dam holds back the huge Stanley Reservoir. Inside the powerhouse, we will see the penstocks, the turbines and the generators, and learn how falling water turns turbines to produce electricity. We will also observe the control room, where operators monitor water level, flow and power output on screens, and the spillway gates used to release extra water safely.

For lunch, we will try local dishes such as hot rice with sambar, fish fry from the river and fresh mango juice, which make the trip a cultural experience as well. In the evening, we will walk through the nearby park and enjoy the green landscape and the calm reservoir.

The visit will help us understand renewable energy, water management and the role of civil, mechanical and electrical engineering in one project. We will follow all safety rules, wear helmets where needed, and stay with our group at all times.

Question 2: Technical study tour to an integrated solar and wind hybrid energy farm

A Technical Study Tour to a Solar and Wind Hybrid Energy Farm

We plan a two-day study tour to the Muppandal wind farm region near Kanyakumari, where wind turbines and solar panels work together. Thirty students and two faculty members will take the overnight train from Chennai to Nagercoil, then travel about one hour by a hired van to the farm. We will stay one night in a budget lodge nearby, and return by train on the second night.

On arrival, a site engineer will give a short briefing on how the hybrid plant works. The wind turbines turn on tall towers, and their blades rotate a generator inside the nacelle. We will watch how the strong coastal wind drives them, and how the blades adjust their angle with the wind speed. In the solar section, we will observe rows of photovoltaic panels, inverters and transformers that feed power into the grid. The control room will show how the two sources balance each other: solar power peaks in the afternoon, whereas wind is strong in the evening and

night.

For food, we will taste Nanjil Nadu specialities such as fish curry with rice, banana chips, avial and fresh coconut sweets. Eating in small local hotels along the route will also help us understand the region's culture.

The tour will show us the practical side of renewable energy, and we will follow safety rules by wearing helmets, keeping away from the turbine bases and obeying the engineers' instructions.

Question 3: Technical field visit to an automated seawater reverse osmosis desalination plant

A Technical Field Visit to a Seawater Reverse Osmosis Desalination Plant

Our class is planning a half-day field visit to a seawater reverse osmosis desalination plant on the East Coast Road near Chennai, such as the Nemmeli plant. About fifty students and three faculty members will travel in two college buses, leaving at 8:00 a.m. and reaching the plant in about ninety minutes. We will carry ID cards and wear closed shoes for safety.

At the plant, the engineers will first explain the process. We will see the intake pumping station, which draws seawater through long pipes under the sea bed. The water then passes through pre-treatment filters that remove sand and suspended particles. Next, high-pressure pumps push the water through the membrane filtration racks, where the salt is separated and fresh water is produced. We will also see the brine discharge outlet, where the concentrated salt water is carefully diluted and released back into the sea to protect marine life. The control room shows the whole process through automated screens.

After the visit, we will have lunch at a local seaside restaurant. The coastal menu includes fish fry, prawn masala, crab curry with steamed rice and filter coffee, which gives us a taste of the Chennai coastal culture.

The visit will help us see how engineering solves water scarcity, how energy is used in the process, and why environmental care in brine disposal is important.

Worksheet 2 – Topic 4: Formal Letters of Request

Question 1: Letter to the Head of the Computer Science Department (lab access for a hackathon)

Inba Ilakiyan
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Head of the Department
Department of Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for permission and lab access after college hours to prepare for a National AI Hackathon

Respected Sir/Madam,

I am writing on behalf of my project team of four first-year students. We have registered for a National AI Hackathon to be held on [date], and we respectfully request permission to use the computer laboratory after college hours.

The objective of our project is to build a simple AI-based solution that solves a real campus problem, using Python and machine learning. We need about three weeks to design, train and test the model, and our home systems are not powerful enough for this.

We require access to the lab between 4:30 p.m. and 7:30 p.m. on weekdays, with Python, Jupyter Notebook, TensorFlow or scikit-learn, Git and a stable internet connection. We promise to follow all lab rules, maintain the equipment carefully, and leave the lab clean and switched off after use. We will also work under the supervision of a faculty member if required.

We would be very grateful if you could grant us this permission.

Thank you for your kind consideration.

Yours obediently,
Inba Ilakiyan
(On behalf of the project team)

Question 2: Letter to the Plant Manager (industrial plant visit)

Inba Ilakiyan
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Plant Manager
[Name of Factory], Industrial Robotics and Automation Division
[Address]

Subject: Request for permission to conduct an industrial visit for 50 first-year engineering students

Respected Sir/Madam,

I am a first-year engineering student of St. Joseph's College of Engineering, Chennai. On behalf of my class, I request your permission to visit your robotics and automation factory with fifty students and two faculty members on a convenient date in [month].

The intended learning outcomes of the visit are to understand how robots and automated systems work on a real production line, to see how sensors, controllers and robotic arms are combined, and to learn about industrial safety practices and quality control. The visit will also help us connect our classroom learning with practical engineering and inspire us to think about careers in automation.

We will arrive in a group, carry our ID cards, follow all safety rules, and obey your staff's instructions throughout. We would be glad if a short briefing and a guided tour of about two hours could be arranged.

I would be grateful if you could kindly grant us this opportunity. I look forward to your reply.

Yours faithfully,
Inba Ilakiyan

Question 3: Letter to the Head of the Mechanical Engineering Department (machine shop access for an electric go-kart)

Inba Ilakiyan
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Head of the Department
Department of Mechanical Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for permission and machine shop access after college hours to build an electric go-kart

Respected Sir/Madam,

Our student team has been selected to take part in an inter-collegiate design challenge to be held on [date], where we must build and test an energy-efficient electric go-kart. We therefore request permission to use the machine shop after regular college hours.

We need the workshop to cut and weld the chassis, fit the motor, battery and braking system, and test the kart on the campus ground. We estimate that we will need about four weeks, working from 4:30 p.m. to 7:30 p.m. on weekdays and during part of the weekends.

We will work only under the supervision of the workshop instructor, wear safety glasses, gloves and shoes at all times, and follow all machine safety rules. We will take care of the tools and keep the shop clean after use.

We would be thankful if you could grant us this permission and allow us to bring honour to the college.

Thank you for your kind support.

Yours obediently,
Inba Ilakiyan
(On behalf of the team)

Question 4: Letter to the Principal / Dean of Student Affairs (sponsorship and duty leave)

Inba Ilakiyan
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Principal / Dean of Student Affairs
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for financial sponsorship and duty leave to present a drone surveillance prototype at the National Defense Innovation Expo

Respected Sir/Madam,

I am happy to inform you that a five-member multidisciplinary team of our college has been selected to present a drone surveillance prototype at the National Defense Innovation Expo, to be held on [dates] at [venue]. We respectfully request financial sponsorship and duty leave for this event.

Our prototype uses a camera-equipped drone with image processing to monitor sensitive areas in real time. Presenting it at a national level will give us valuable feedback from experts and will bring recognition to our college.

We request sponsorship for travel, accommodation, registration and prototype transport, which we estimate at ₹[amount], and duty leave for [number] days so that we do not lose attendance. We will make up for the missed classes and submit a report after the event.

I would be grateful for your kind support and approval.

Thank you.

Yours obediently,
Inba Ilakiyan
(On behalf of the team)

Worksheet 2 – Topic 5: Compare and Contrast Descriptive Essays

Question 1: Electric Vehicles versus Internal Combustion Engine Vehicles

Electric Vehicles versus Conventional Internal Combustion Engine Vehicles

Electric vehicles (EVs) and internal combustion engine (ICE) vehicles are two different answers to the same need, which is transport. Each has clear positives and negatives.

Consider battery life first. An EV battery usually lasts eight to ten years, but replacing it is very expensive. In contrast, an ICE vehicle has no battery problem, though its engine needs regular servicing. As for the manufacturing footprint, EVs create more emissions during production because of battery mining and processing, whereas ICE vehicles have a lighter manufacturing footprint. However, over its lifetime an EV

produces far lower emissions, especially when the electricity comes from renewable sources.

Operating costs also differ. EVs have fewer moving parts, so they need less maintenance, and electricity per kilometre is cheaper than petrol or diesel. On the other hand, ICE vehicles cost less to buy, and their repair shops are available everywhere. Fuel prices, however, keep rising and make them costly to run.

Charging network feasibility is the biggest challenge for EVs. Charging stations are still limited, especially on highways and in small towns, and charging takes much longer than refuelling. ICE vehicles, conversely, can be refuelled in minutes at petrol stations spread across the country.

In conclusion, EVs are cleaner and cheaper to run, whereas ICE vehicles are cheaper to buy and more convenient at present. As battery prices fall and charging networks grow, EVs are likely to become the better choice for the future.

Question 2: Remote/Virtual Internships versus On-Site Internships

Remote Internships versus On-Site Internships for First-Year Engineering Students

Internships help early-stage engineering students connect classroom theory with real work. Today, students can choose between remote (virtual) and on-site (physical) internships, and each has advantages and disadvantages.

Hands-on skill acquisition is the biggest difference. On-site internships give direct access to machines, laboratories and real equipment, so students learn practical skills such as wiring, fabrication and testing. Remote internships, in contrast, mostly involve software, simulations and online tasks, which are useful for coding and design but cannot replace physical practice.

In terms of schedule flexibility, remote internships win clearly. Students can work from home, save travel time and cost, and manage studies alongside the internship. On-site internships, on the other hand, demand fixed hours and daily travel, which can be tiring during the semester.

Mentorship is stronger on-site, because supervisors can guide students directly and correct mistakes immediately. Online mentors, however, are often busy and communicate only through scheduled calls or messages, so doubts may stay unresolved for days.

Professional networking also favours on-site internships, as students meet colleagues, build relationships and learn workplace culture naturally. Remote interns may never meet their team face to face, although online platforms still allow them to connect with professionals from different cities.

In conclusion, remote internships are better for flexibility and software skills, whereas on-site internships are better for practical learning, mentorship and networking. A first-year student should ideally begin with a remote internship and move to an on-site one later.

Question 3: Cloud Computing Infrastructure versus On-Premise Data Centers

Cloud Computing versus On-Premise Physical Data Centers for Engineering Enterprises

Engineering enterprises must decide whether to run their systems on the cloud or in their own physical data centers. Both choices have strengths and weaknesses.

Upfront capital expenditure is the clearest difference. Cloud services need almost no initial investment, because companies simply pay for what they use. On-premise data centers, in contrast, require heavy spending on servers, cooling systems, power backup and buildings before any work begins.

As for recurring overheads, cloud bills can grow quickly if usage is not monitored, whereas an on-premise setup has predictable costs once it is built, although it still needs staff, electricity and maintenance. Therefore, cloud suits changing workloads, while on-premise suits stable and long-term demand.

Workload scalability strongly favours the cloud. Companies can add computing power in minutes during simulations or product launches and reduce it afterwards. On the other hand, on-premise systems have fixed capacity, so expanding them takes weeks or months and leaves hardware idle during quiet periods.

Data security and compliance are more complex. On-premise data centers give the company full physical control, which is important for defence, healthcare or confidential design data. Cloud providers offer strong security tools and certifications, but the company depends on a third party and must follow data-location laws

carefully.

In conclusion, the cloud is better for flexibility, speed and low starting cost, while on-premise infrastructure is better for control and predictable long-term expenses. Many enterprises now choose a hybrid model that combines the advantages of both.

Question 4: 3D Printing versus Traditional CNC Milling

3D Printing (Additive Manufacturing) versus CNC Milling (Subtractive Manufacturing) in Prototype Fabrication

In prototype fabrication, 3D printing and CNC milling follow opposite principles. A 3D printer builds a part layer by layer, whereas a CNC machine cuts material away from a solid block. Each method has positives and negatives.

Material wastage is the first point of contrast. 3D printing uses only the material needed for the part, so wastage is low. CNC milling, on the other hand, removes a large amount of material as chips, which increases the cost of expensive metals.

When it comes to production speed, the picture changes with batch size. For single prototypes and small batches, 3D printing is faster because it needs no tooling or setup. For larger batch runs, however, CNC milling is quicker, as it can produce identical parts continuously at high speed.

Surface finish and accuracy favour CNC milling. It produces smooth surfaces and very tight tolerances, so parts often need no extra work. In contrast, 3D printed parts often show layer lines and need sanding or polishing, and their accuracy is lower.

Geometric complexity is the strength of 3D printing. It can easily create hollow parts, internal channels and lattice structures that are impossible to cut with a rotating tool. CNC milling is limited by tool access and the direction of cutting.

In conclusion, 3D printing is ideal for complex, low-volume prototypes with little waste, whereas CNC milling is better for accurate, smooth and larger-batch parts. Engineers often use both methods together to get the best result.

Worksheet 2 – Topic 6: Personal Email Writing

Question 1: Email to a high school mentor/teacher about the first semester

To: [teacher's email address]

Subject: My first semester at engineering college

Dear Sir/Madam,

I hope you are doing well. I wanted to write to you because your guidance in school gave me the confidence to start my engineering journey, and I would like to share how my first semester has been.

The biggest change was the move to independent campus life. Nobody reminds us about deadlines now, so I had to learn to manage my own time, plan my study hours and take responsibility for my work. It was difficult at first, but I am getting used to it.

In our laboratory sessions, we worked on our first small projects, such as basic programs in Python and C and simple circuits. Seeing my own code run correctly gave me a great sense of satisfaction, and I realised how different practical learning is from reading theory.

For the end-semester examinations, I have made a timetable that gives equal time to every subject. I revise the notes every weekend, solve previous question papers and practise the lab experiments so that I do not forget the procedures.

Thank you for everything you taught me. I will be very happy to hear from you and to receive any advice for the coming semesters.

With respect and regards,
Inba Ilakiyan

Question 2: Email to a high school science teacher about the first term

To: [science teacher's email address]

Subject: Reflections on my first term in engineering

Dear Sir/Madam,

I hope you are keeping well. It has been a few months since I joined engineering college, and I wanted to share my first-term experience with you, since you were the one who made science exciting for me.

The first thing I noticed was how different the hands-on lab coursework is. In school, we mostly saw experiments in a few periods, but here we handle components, tools and software ourselves. I made mistakes in the beginning, but each mistake taught me something and I am becoming more careful and confident.

I also took part in my first technical club activity and a small hackathon. Working with a team to build a solution within a limited time was challenging and thrilling, and I learned a lot about teamwork and debugging from my seniors.

For the upcoming university theory and practical exams, my strategy is simple. I will finish revising each unit at least two weeks before the exams, practise numerical problems daily, and repeat the lab experiments so that I can answer viva questions without fear.

I would be grateful for any advice you may have. Please take care.

Yours sincerely,
Inba Ilakiyan

Question 3: Email to an elder sibling/alumnus seeking advice

To: [sibling's or alumnus's email address]

Subject: Need your advice on balancing studies and technical clubs

Dear [Name],

How are you? I hope college and work are going well. I am writing because I need your advice on something I am struggling with.

My day starts at 6:30 a.m., and I attend classes until 4:30 p.m. After that, I go to my technical club meetings for about two hours, and in the evening I try to study and finish my assignments. By the time I am free, I am usually very tired.

My biggest challenge is managing the lab record submissions. Because the records must be completed every week, they pile up, and I often finish them at the last minute, which affects the quality.

Since you have handled the same situation, could you please tell me how you balanced your academics with the clubs? I would also like your tips for semester exam revision, such as how to plan the timetable, which subjects to start with, and how to use previous question papers.

I am looking forward to your reply. Thank you in advance, and please take care.

Your loving brother/sister,
Inba

Worksheet 2 – Topic 7: Travel Planning and Budgetary Itineraries

Question 1: 4-day tour to an aerospace and satellite launch station

A 4-Day Student Technical Tour to an Aerospace and Satellite Launch Station

Our department plans a four-day tour to a satellite launch station for forty students and two faculty members. We will travel by a chartered train coach and local bus, so that the cost is reasonable.

Budget (per student): train fare ₹1,200, local bus ₹400, accommodation for three nights ₹1,500, food ₹1,200, entry and miscellaneous charges ₹400. The total is ₹4,700, which gives about ₹1,88,000 for the full group of forty.